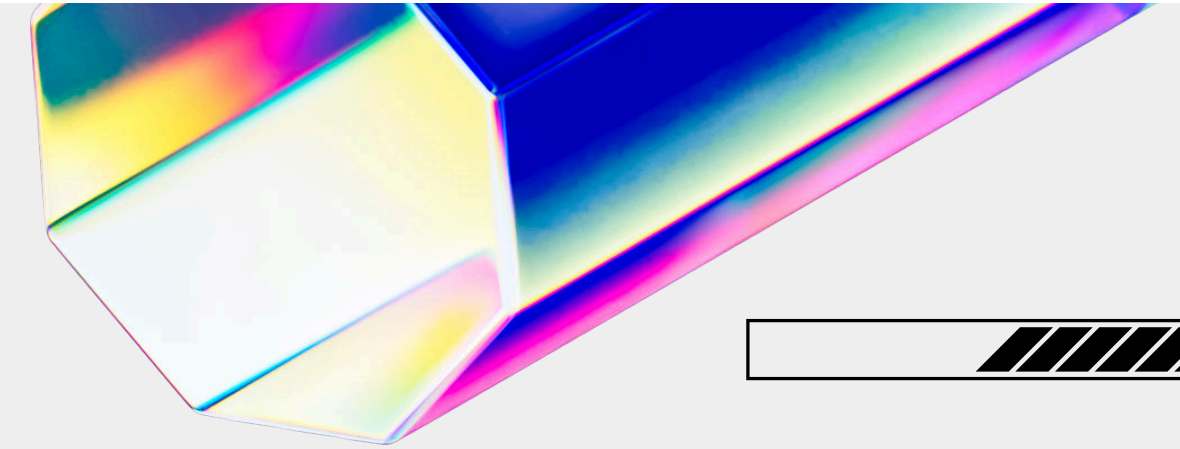


PROJET YAS'INNOV



STOCKBOT

PRESENTED BY :
MAHMOUDI ASSAAD



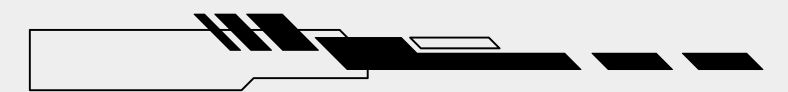
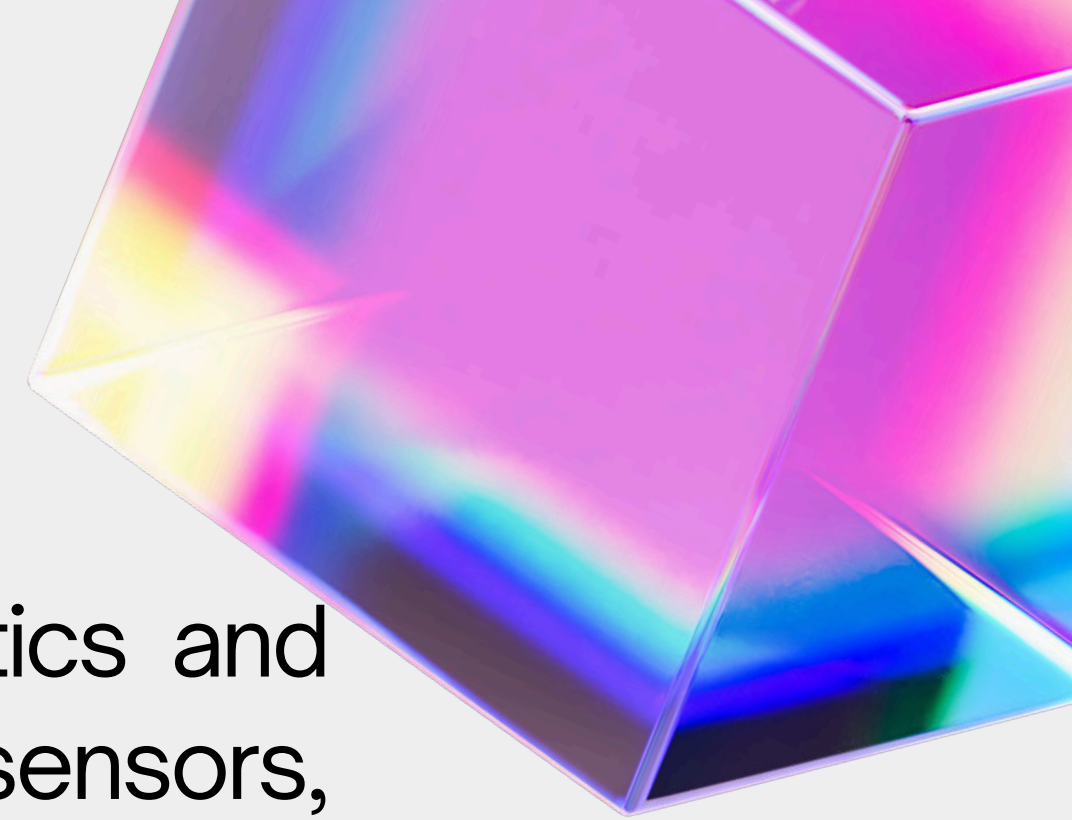
Projet financé par
l'Union européenne



INTRODUCTION

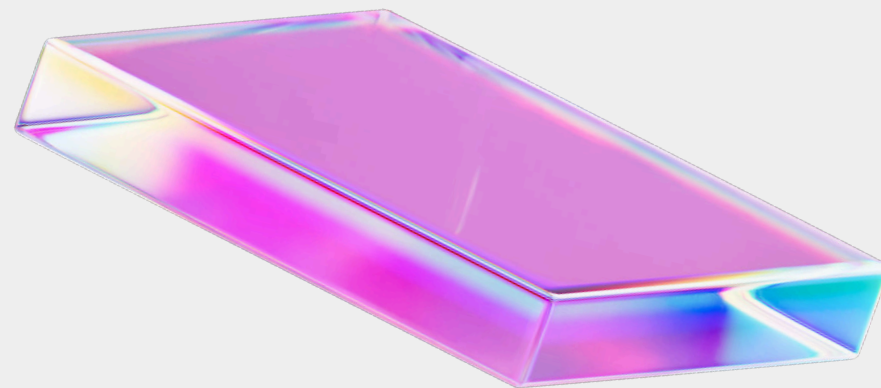
I am an engineering student passionate about robotics and automation. I enjoy working with embedded systems, sensors, and intelligent machines. Through my studies and club activities, I have gained experience in designing and building practical robotic solutions.

With the STOCKBOT project, I aim to use my skills to solve real problems in stock management by creating an autonomous robot that improves efficiency and reduces human effort.



PROBLEM

- frequent human errors in inventory records,
- loss of time searching for items,
- poor traceability of products,
- and heavy workload for employees.



IMPORTANCE

- save time,
- reduce operational costs,
- improve accuracy and reliability,
- and increase productivity.



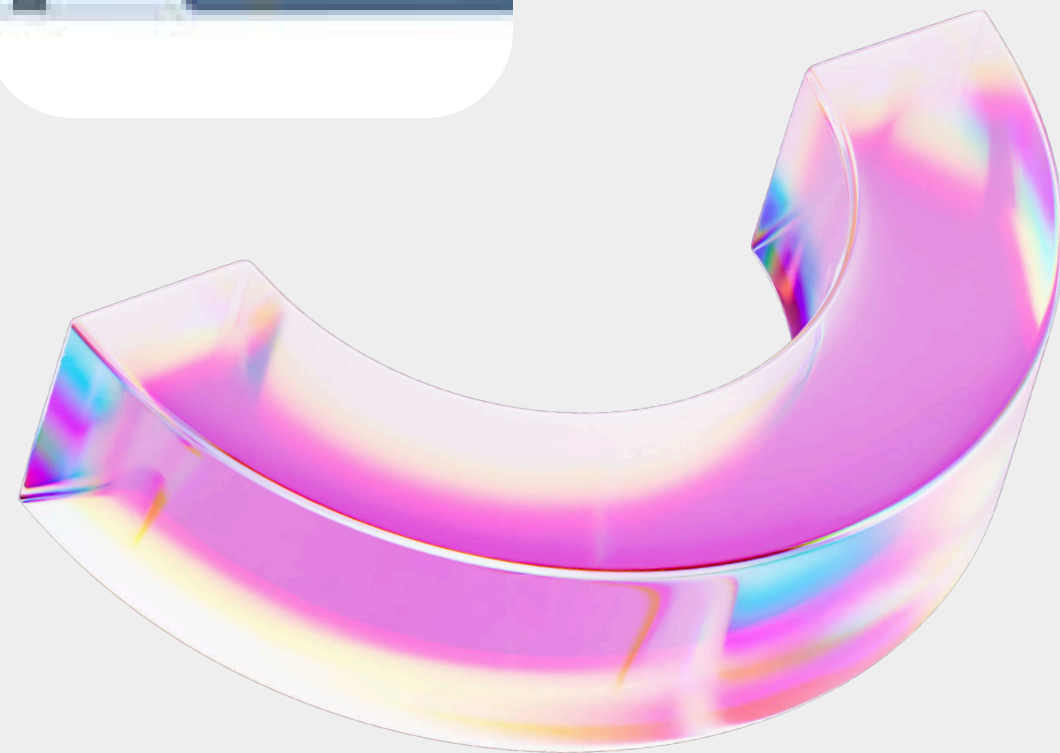
PROJECT DESCRIPTION

STOCKBOT is an autonomous mobile robot designed to automate stock management by moving independently, identifying items, and updating inventory in real time.

PRODUCT / SERVICE

An intelligent robot capable of:

- *Navigating inside storage areas,*
- *Scanning QR codes or RFID tags,*
- *Transporting light objects,*
- *Communicating with a digital inventory system.*



8) Key Partners - University laboratories and engineering schools (e.g., ENIG). - Local industrial companies for pilot testing. - Electronics component suppliers. - Robotics clubs and communities. - Supporting associations (e.g., FIDEL Metouia Association).	6) Key Activities - Robot hardware design and prototyping. - Software programming and embedded AI development. - Rigorous testing and validation. - Continuous product improvement. - Integration with clients' existing inventory management systems.	2) Value Proposition - Time saving through automation. - Reduction of human errors in inventory counting. - Automation of repetitive and manual stock-taking tasks. - Better product traceability in real-time. - Reduction of manual labor costs.	4) Customer Relationships - Technical support and troubleshooting. - After-installation assistance and setup. - User training sessions. - Regular software and feature updates. - Custom adaptation to specific client needs.	1) Customer Segments - Industrial warehouses - Technical workshops - University laboratories - Stores and logistics centers - Scientific clubs and fablabs
	7) Key Resources - Electronic components (microcontrollers, sensors, motors). - Programming software and AI algorithms. - Physical resources (robot chassis, batteries). - Team expertise in robotics and AI. - Manufacturing tools (3D printers, workshop).		3) Channels - Live demonstrations and trade shows. - Presentations in academic and innovation competitions. - Social media (LinkedIn, Instagram). - Direct partnerships with local companies. - Collaboration with university networks and robotics clubs.	
9) Cost Structure - Cost of robotic materials and components. - Sensors and electronic boards. - Batteries and mechanical parts. - Costs for 3D printing and fabrication. - Expenses related to testing, prototyping, and R&D.			5) Revenue Streams - Sale of the STOCKBOT robot unit. - Recurring revenue from maintenance services. - Fees for customization and special features. - Annual technical support contracts.	



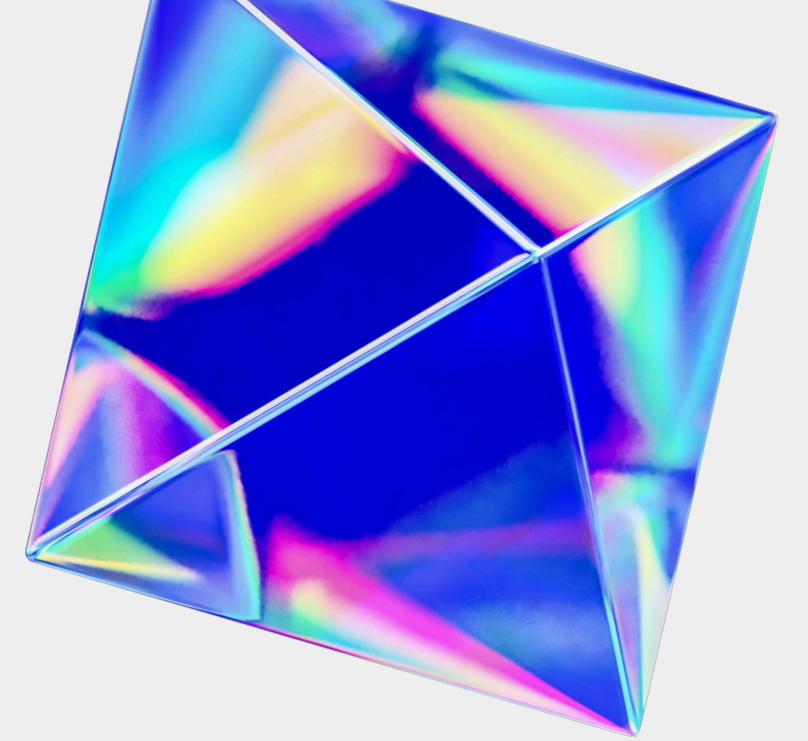
CUSTOMER SEGMENTS (TARGET MARKET)

Industrial warehouses
Technical workshops
University laboratories
Stores and logistics centers
Scientific clubs and fablabs



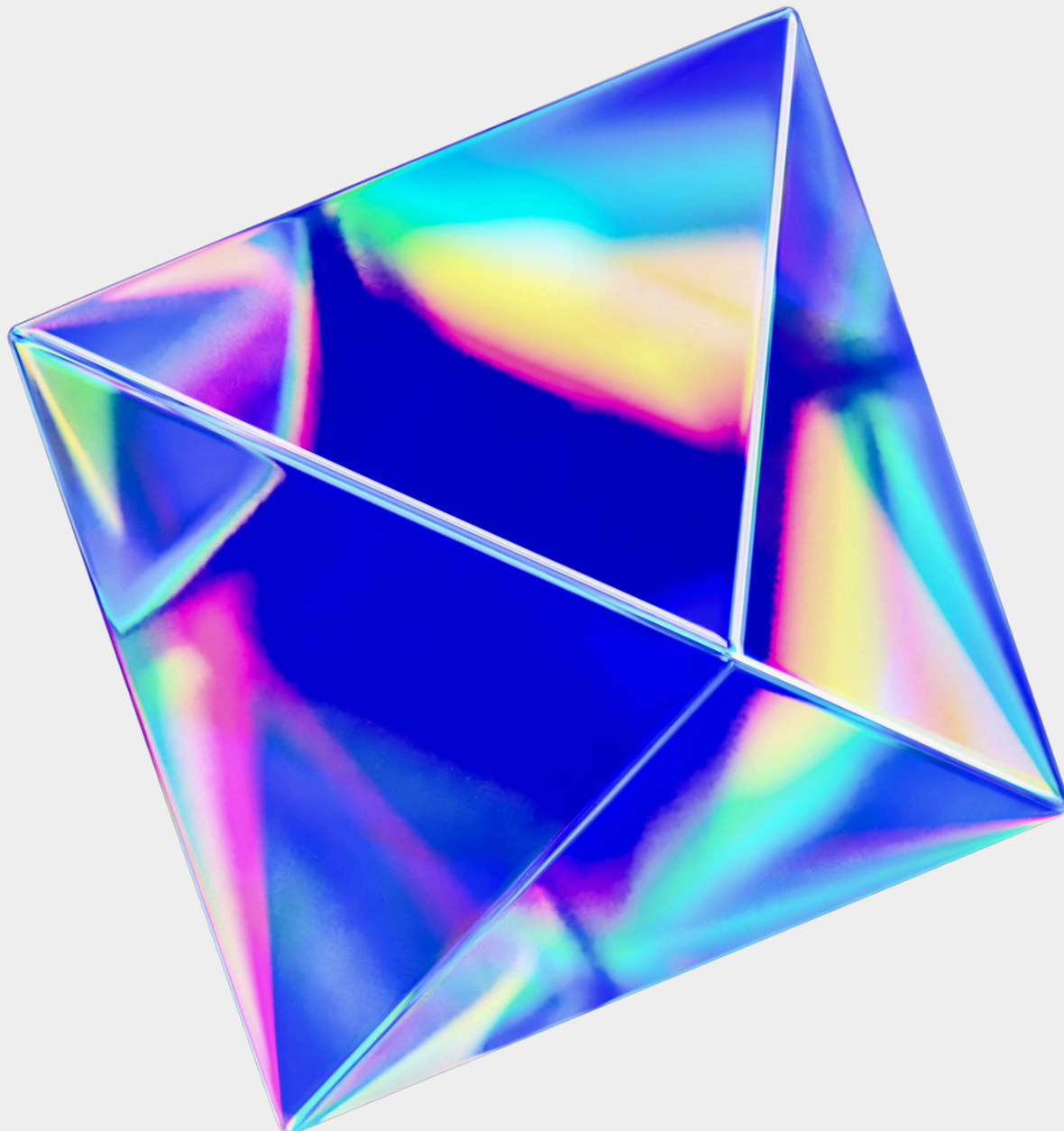


VALUE PROPOSITION



STOCKBOT provides:

- Time saving
- Reduction of human errors
- Automation of repetitive tasks
- Better product traceability
- Less manual labor



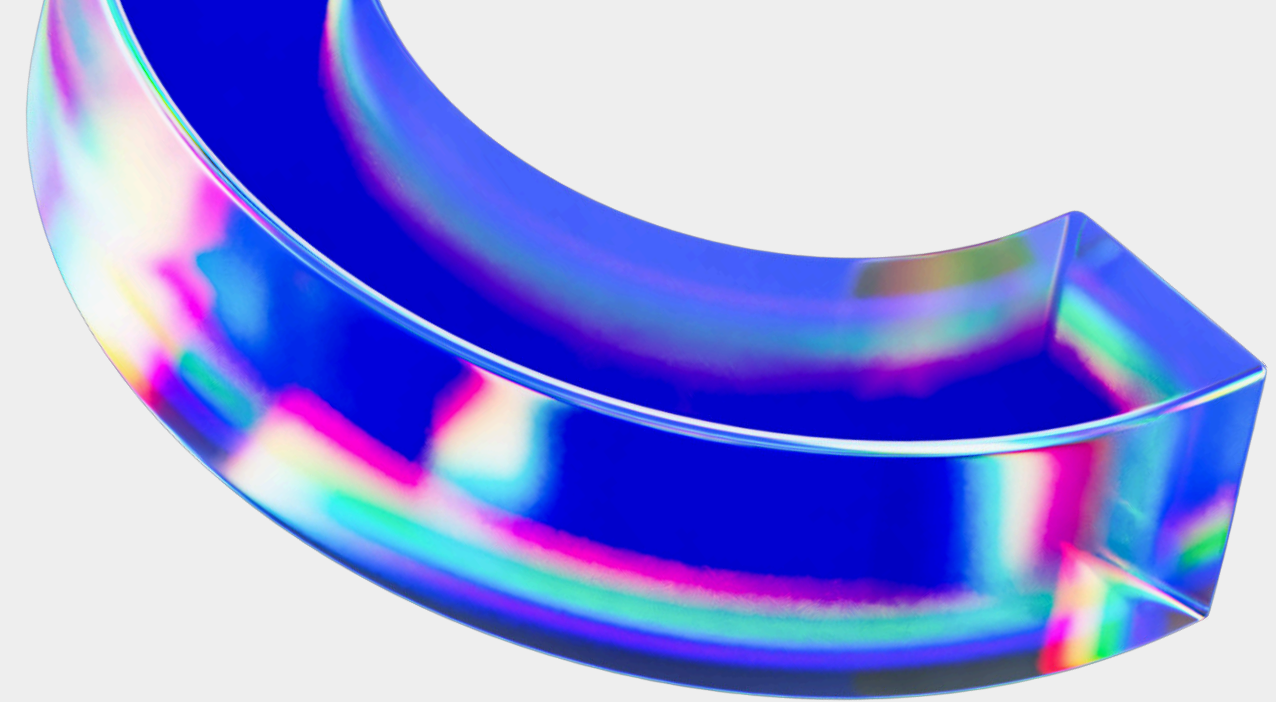
COMMUNICATION AND DISTRIBUTION

Communication Channels:

- 1. Live demonstrations of the robot*
- 2. Presentations at competitions and events*
- 3. Social media (LinkedIn, Facebook, Instagram)*
- 4. Collaboration with university and robotics clubs*
- 5. Direct contact with companies*

Distribution Channels:

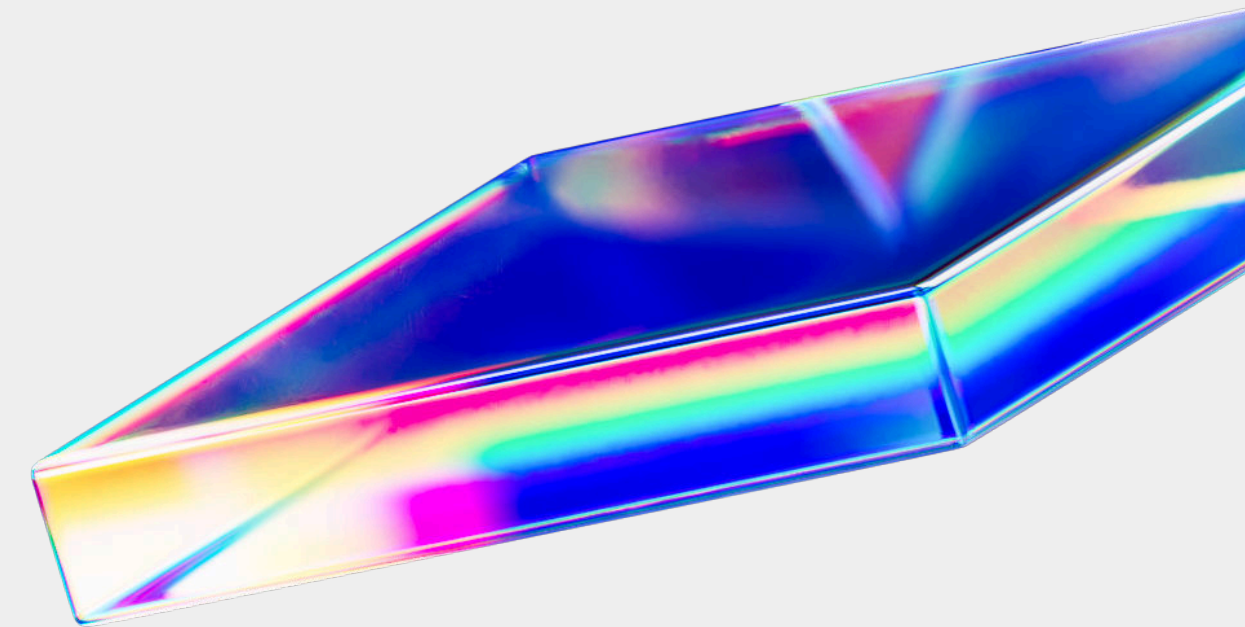
- 1. Direct sale to companies and laboratories*
- 2. Custom orders based on client needs*
- 3. Partnerships with local industries*
- 4. University and club collaborations*





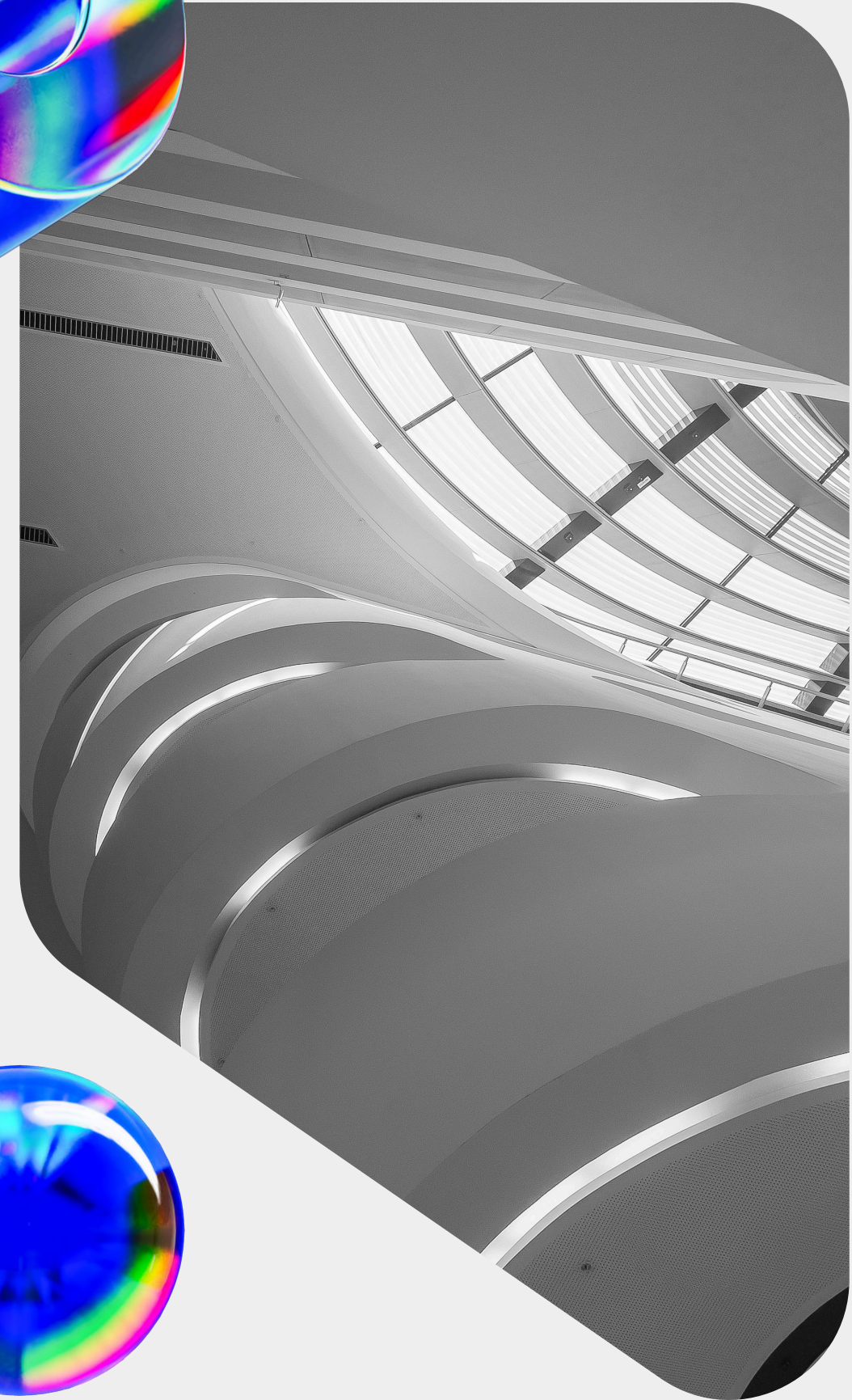
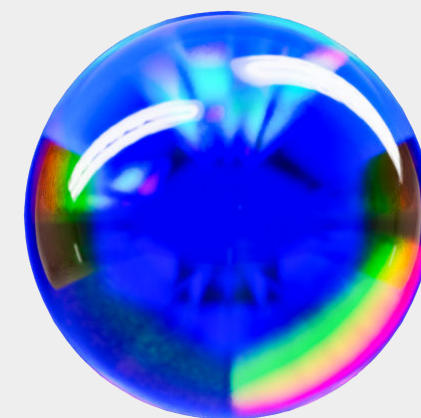
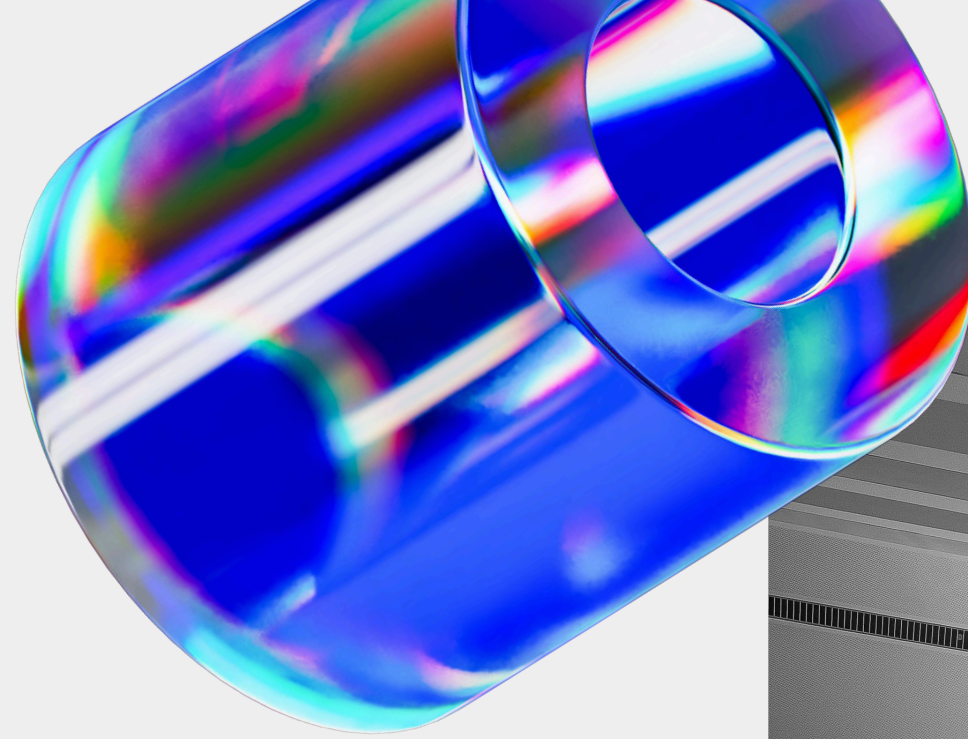
- *Technical support*
- *After-installation assistance*
- *User training*
- *System updates*
- *Custom adaptation to client needs*

CUSTOMER RELATIONSHIP



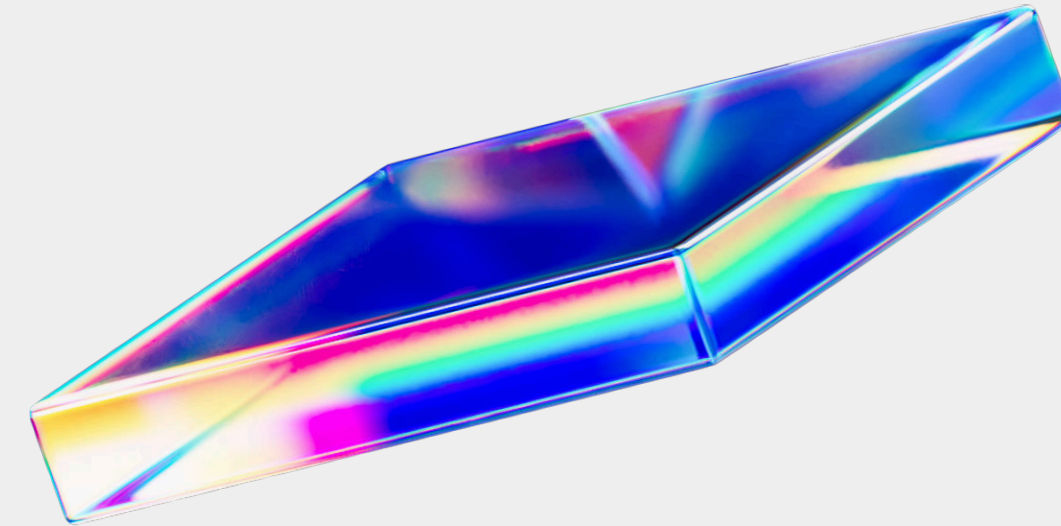
REVENUE STREAMS

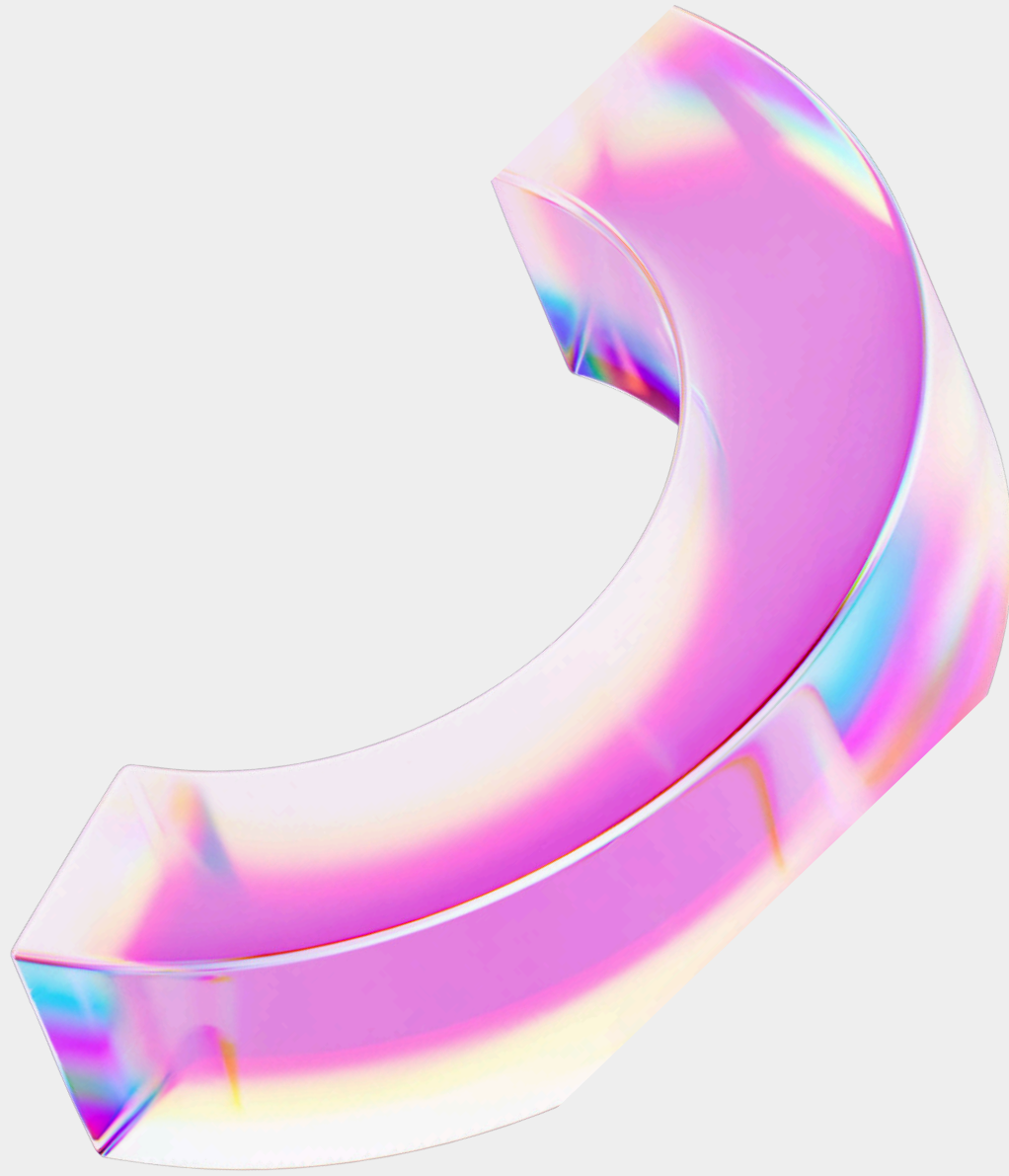
- SELLING THE ROBOT TO COMPANIES
- MAINTENANCE SERVICES
- CUSTOMIZATION BASED ON CLIENT NEEDS
- TECHNICAL SUPPORT CONTRACTS



KEY ACTIVITIES

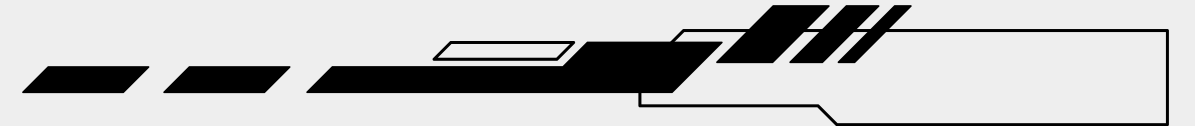
- ROBOT DESIGN
- PROGRAMMING AND EMBEDDED INTELLIGENCE
- TESTING AND VALIDATION
- CONTINUOUS IMPROVEMENT
- INTEGRATION WITH EXISTING SYSTEMS





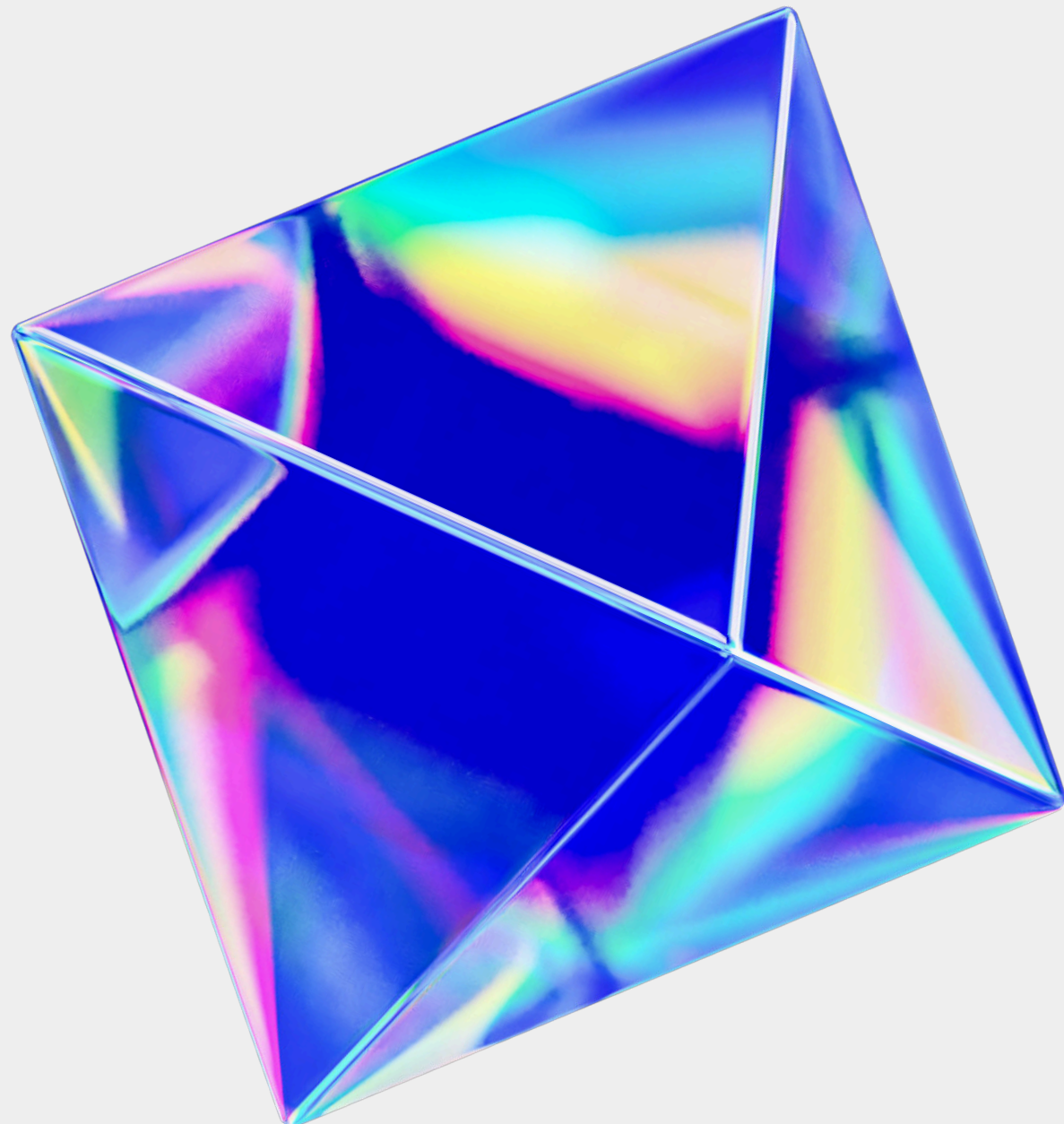
KEY RESOURCES

- Electronic components (Arduino/ESP32, sensors, motors)
- Programming software
- Robot chassis and battery
- Robotics and AI skills
- Manufacturing tools





KEY PARTNERS



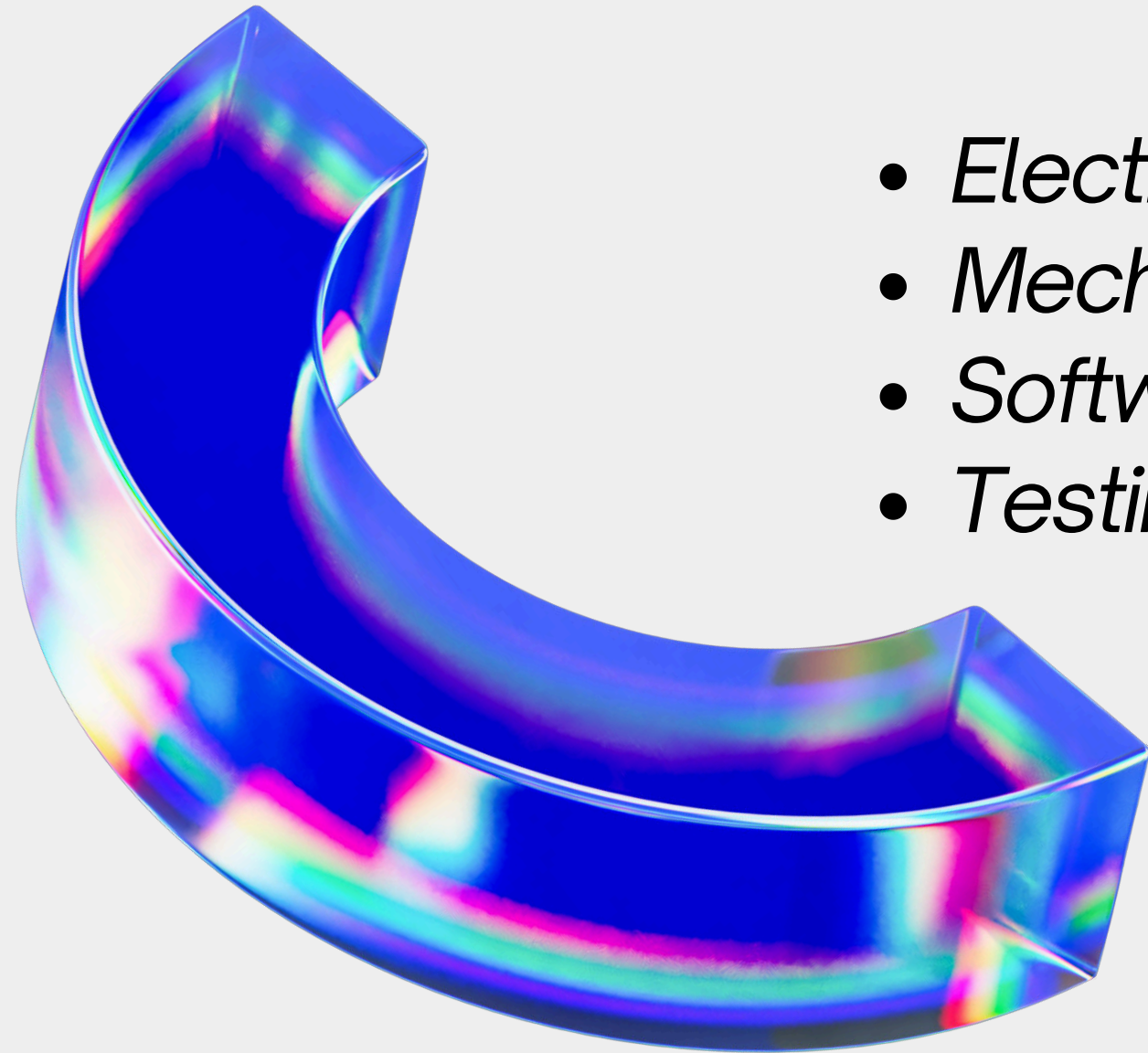
Primary Partners (Tunisian & Relevant)

- ENIG (University of Gabès) — technical support, labs, mentoring
- ENIG Robotics Club — prototyping, testing, teamwork
- ARS Tunisia (Automation & Robotics) — industrial robotics expertise
- Scantech Tunisia (RFID & Logistics) — product identification systems
- AIOT Tunisia (IoT Solutions) — sensors & real-time data
- Local Warehouses / SMEs — testing site and real use cases

Supporting Partners

- Tunisian FabLabs / MakerSpaces — 3D printing & fabrication
 - FIDEL Metouia Association — project framework (YAS'Innov)
-

COST STRUCTURE

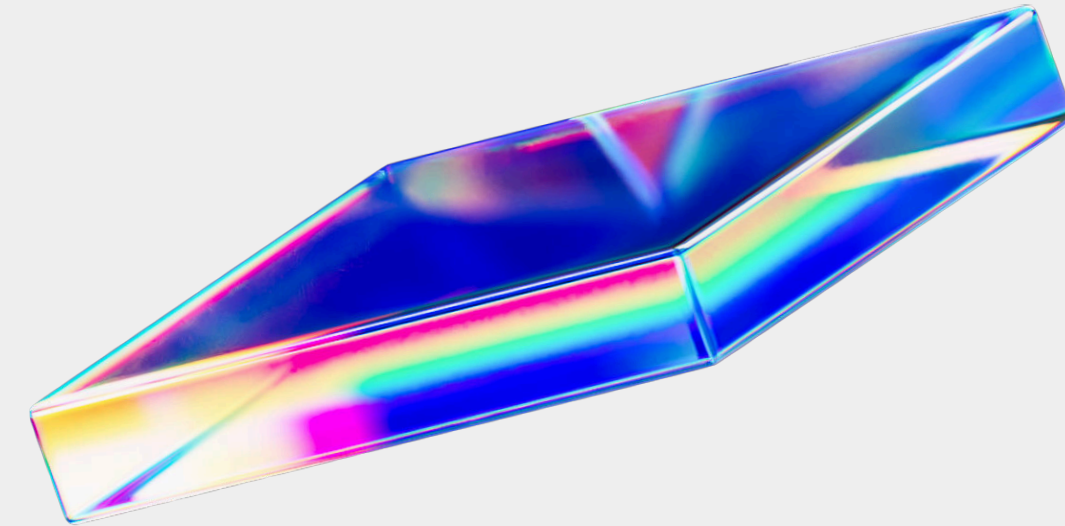


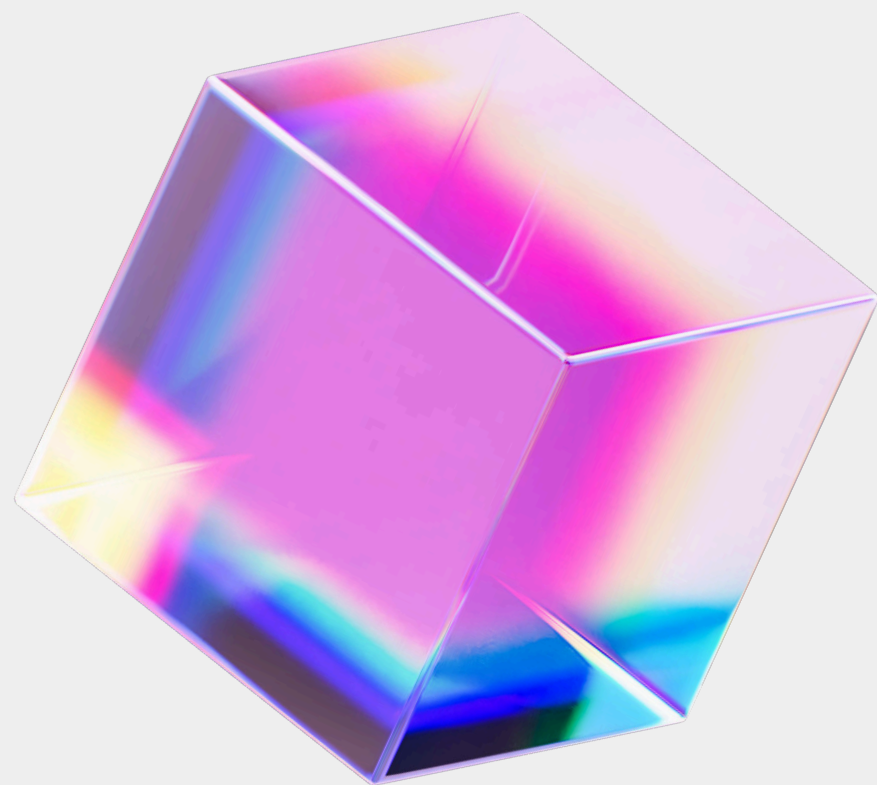
- *Electronics: ESP32, sensors, motors, battery*
- *Mechanical: 3D printing, chassis, robotic materials*
- *Software: Development & tools*
- *Testing & Maintenance*



SOCIAL AND ENVIRONMENTAL IMPACT

- REDUCTION OF WASTE
- BETTER RESOURCE MANAGEMENT
- FEWER INVENTORY LOSSES
- CONTRIBUTION TO INDUSTRY 4.0
- DEVELOPMENT OF LOCAL ROBOTICS SKILLS





THANK
YOU

